

MATH I (Fall 2026)

WEEK 2: Practical Session Problems (on Functions)

Exercise 1: Relations and functions

Determine whether each relation defines y as a function of x . Explain your answer.

a. $y = 3x - 2$,

b. $x^2 + y^2 = 9$,

Where appropriate, solve for y and apply the vertical-line test.

Exercise 2: Function notation

Let

$$f(x) = 2x^2 - 3x + 1.$$

Calculate and simplify:

$$f(0), \quad f(-2), \quad f(a), \quad f(a + h),$$

$$\frac{f(a + h) - f(a)}{h}, \quad h \neq 0.$$

Exercise 3: Domain

Find the natural domain of each real-valued function:

a. $f(x) = \frac{x + 1}{x^2 - 9}$,

b. $g(x) = \sqrt{5 - 2x}$,

Express each domain using interval notation.

Exercise 4: Composite functions

Let

$$f(x) = 2x - 3, \quad g(x) = x^2 + 1.$$

Find:

$$(f \circ g)(x), \quad (g \circ f)(x),$$

Are $f \circ g$ and $g \circ f$ equal? Explain why function composition is generally not commutative.

Exercise 5: Symmetry

Determine whether each function is even, odd or neither:

$$|x| + 1, \quad \frac{x}{x^2 + 1}.$$

Verify your conclusions algebraically and graphically.

Exercise 6: Identify the function class

Classify each function as linear, polynomial, rational, algebraic or transcendental. A function may belong to more than one category.

$$\frac{x + 1}{x^2 + 4}, \quad \sqrt{x^2 + 1}$$

Exercise 7: Linear engineering model

The temperature inside an industrial chamber increases linearly from 20°C at $t = 0$ to 80°C at $t = 15$ minutes.

1. Find a function $T(t)$ describing the temperature.
 2. Interpret its slope and vertical intercept.
 3. Find $T(8)$.
 4. At what time does the chamber reach 60°C?
 5. State a physically reasonable domain.
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Exercise 8: Polynomial behaviour

For

$$p(x) = x^3 - 4x$$

determine:

- domain;
 - intercepts;
 - symmetry, if any;
 - long-run or end behaviour;
 - locations where the graph is undefined.
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Exercise 9: Piecewise electricity tariff

An electricity provider charges according to

$$C(E) = \begin{cases} 0.10E, & 0 \leq E \leq 200, \\ 20 + 0.15(E - 200), & E > 200, \end{cases}$$

where E is monthly energy consumption in kWh and $C(E)$ is the cost.

1. Find $C(150)$, $C(200)$ and $C(350)$.
 2. Sketch the graph.
 3. Is the cost model continuous at $E = 200$?
 4. Explain the meaning of each slope.
 5. Suggest one limitation of the model.
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Exercise 10: Finding inverse functions

Find the inverse of the following function and state the domain of the function and its inverse:

$$f(x) = 4x - 7,$$

Verify your answer using composition:

$$f(f^{-1}(x)) = x.$$

Exercise 11: Exponential growth and decay

For each model, decide whether it represents growth or decay and identify its initial value and growth or decay factor:

$$A(t) = 500(1.08)^t,$$

$$B(t) = 1200(0.85)^t,$$

$$C(t) = 40e^{0.03t},$$

$$D(t) = 200e^{-0.4t}.$$

Exercise 12: Solve exponential equations

Solve:

$$2^{x+1} = 16,$$

Exercise 13: Battery discharge

The stored energy in a battery is modeled by

$$E(t) = E_0 e^{-0.025t},$$

where t is measured in hours.

1. What fraction of the initial energy remains after 10 hours?
 2. When will 50% of the initial energy remain?
 3. When will 10% remain?
 4. Explain why the model never predicts exactly zero energy.
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Exercise 14 (if time allows): Population growth

A bacterial population is modeled by

$$P(t) = 800e^{0.35t},$$

where t is measured in hours.

1. Find the initial population.
 2. Find the population after 5 hours.
 3. Determine the doubling time.
 4. Explain whether the model can remain realistic indefinitely.
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